

```
sh@DESKTOP-0TK9CH3: ~  
sh@DESKTOP-0TK9CH3:~$ ps -ef  
UID          PID     PPID  C  STIME TTY          TIME CMD  
root          1         0  0  16:53 ?        00:00:00 /sbin/init  
root          2         1  0  16:53 ?        00:00:00 /init  
root          7         2  0  16:53 ?        00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --  
root         53         1  0  16:53 ?        00:00:02 /usr/lib/systemd/systemd-journald  
root        111         1  0  16:53 ?        00:00:00 /usr/lib/systemd/systemd-udevd  
systemd+    124         1  0  16:53 ?        00:00:00 /usr/lib/systemd/systemd-resolved  
systemd+    125         1  0  16:53 ?        00:00:00 /usr/lib/systemd/systemd-timesyncd  
root        178         1  0  16:53 ?        00:00:00 /usr/sbin/cron -f -P  
message+    179         1  0  16:53 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --syst  
root        186         1  0  16:53 ?        00:00:00 /usr/lib/systemd/systemd-logind  
root        188         1  0  16:53 ?        00:00:00 /usr/libexec/wsl-pro-service -vv  
root        190         1  0  16:53 hvc0     00:00:00 /sbin/agetty -o -p -- \u --noclear --keep-baud - 115200,38400,9600 v  
root        198         1  0  16:53 tty1     00:00:00 /sbin/agetty -o -p -- \u --noclear - linux  
syslog      199         1  0  16:53 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE  
root        206         1  0  16:53 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s  
root        363         2  0  16:53 ?        00:00:00 /init  
root        364        363  0  16:53 ?        00:00:00 /init  
sh          368        364  0  16:53 pts/0    00:00:00 /bin/bash  
root        483         2  0  16:55 pts/1    00:00:00 /bin/login -f  
sh          526         1  0  16:55 ?        00:00:00 /usr/lib/systemd/systemd --user  
sh          527        526  0  16:55 ?        00:00:00 (sd-pam)  
sh          542        483  0  16:55 pts/1    00:00:00 -bash  
root        837         2  0  17:11 ?        00:00:00 /init  
root        839        837  0  17:11 ?        00:00:00 /init  
sh          843        839  0  17:11 pts/2    00:00:00 -bash  
sh          858        843  0  17:11 pts/2    00:00:00 ps -ef  
sh@DESKTOP-0TK9CH3:~$ kill -9 526
```

```
sh@DESKTOP-0TK9CH3:~$ sleep 1000 &
```

```
[1] 923
```

```
sh@DESKTOP-0TK9CH3:~$ ps -ef | grep sleep
```

```
sh          923      898  0 17:18 pts/2    00:00:00 sleep 1000
```

```
sh          925      898  0 17:19 pts/2    00:00:00 grep --color=auto sleep
```

```
sh@DESKTOP-0TK9CH3:~$ ps -ef | grep sleep
```

```
sh          923      898  0 17:18 pts/2    00:00:00 sleep 1000
```

```
sh          931      898  0 17:19 pts/2    00:00:00 grep --color=auto sleep
```

```
sh@DESKTOP-0TK9CH3:~$ pkill sleep
```

```
[1]+  Terminated          sleep 1000
```

```
sh@DESKTOP-0TK9CH3:~$ sleep 1000 &
```

```
[1] 933
```

```
sh@DESKTOP-0TK9CH3:~$ ps -ef | grep sleep
```

```
sh          933      898  0 17:20 pts/2    00:00:00 sleep 1000
```

```
sh          935      898  0 17:20 pts/2    00:00:00 grep --color=auto sleep
```

```
sh@DESKTOP-0TK9CH3:~$ kill 933
```

```
sh@DESKTOP-0TK9CH3:~$ ps -ef | grep sleep
```

```
sh          937      898  0 17:21 pts/2    00:00:00 grep --color=auto sleep
```

```
[1]+  Terminated          sleep 1000
```

```
sh@DESKTOP-0TK9CH3:~$ |
```

```
int main()
{
    int pid = fork();

    if (pid == 0)
    {
        // Child process
        printf("Child Process\n");
        printf("PID : %d\n", getpid());
        printf("PPID : %d\n", getppid());
    }
    else
    {
        // Parent process
        printf("Parent Process\n");
        printf("PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);
    }

    return 0;
}
```

gcc process.c -o process

./process

```
sh@DESKTOP-0TK9CH3:~$ nano process.c
```

```
sh@DESKTOP-0TK9CH3:~$ gcc process.c -o process
```

```
./process
```

```
Parent Process
```

```
PID : 2539
```

```
Child PID : 2540
```

```
Child Process
```

```
PID : 2540
```

```
PPID : 2511
```

```
sh@DESKTOP-0TK9CH3:~$ |
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if (pid > 0)
    {
        // Parent
        printf("Parent exiting\n");
        sleep(2);
    }
    else
    {
        // Child
        sleep(5);
        printf("Child PID   : %d\n", getpid());
        printf("New PPID    : %d\n", getppid());
    }

    return 0;
}
```

```
sh@DESKTOP-0TK9CH3:~$ nano orphan.c
sh@DESKTOP-0TK9CH3:~$ nano orphan.c
sh@DESKTOP-0TK9CH3:~$ gcc orphan.c -o orphan
sh@DESKTOP-0TK9CH3:~$ ./orphan
```

Parent exiting

```
sh@DESKTOP-0TK9CH3:~$
sh@DESKTOP-0TK9CH3:~$ Child PID    : 2509
New PPID      : 895
```

```
sh@DESKTOP-0TK9CH3:~$ |
```

```
#include <stdio.h>
#include <unistd.h>

int main()
{
    int pid = fork();

    if (pid == 0)
    {
        // Child
        printf("Child exiting\n");
    }
    else
    {
        // Parent
        sleep(15);
        printf("Parent still running\n");
    }

    return 0;
}
```



```
sh@DESKTOP-0TK9CH3:~$ ps -e1 | grep Z
```

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
---	---	-----	-----	------	---	-----	----	------	----	-------	-----	------	-----

```
sh@DESKTOP-0TK9CH3:~$ |
```